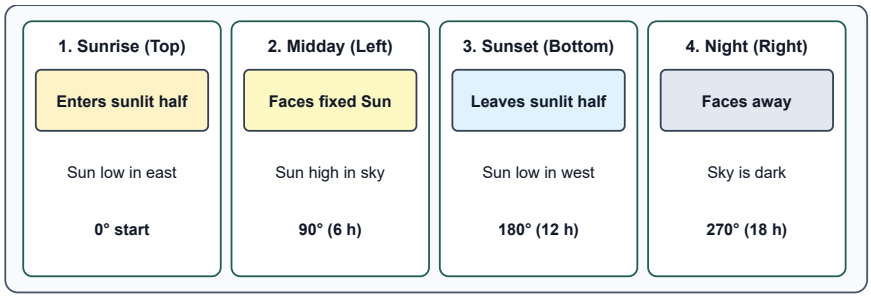
# Reference sheets

## Reference A · One marked place rotating through light and dark (North-Pole view)

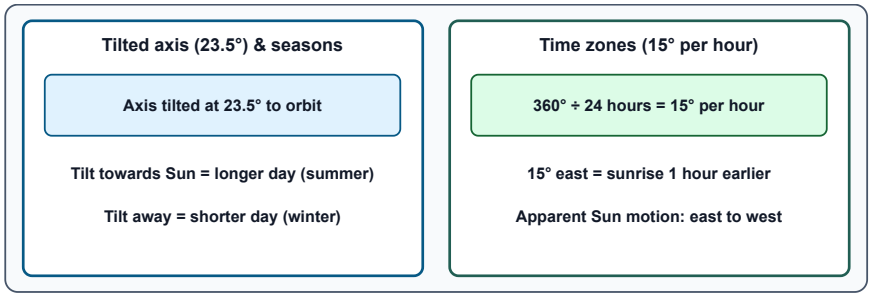



Four positions of a marked place in the North-Pole view with sunlight from the left: 1. Sunrise, 2. Midday, 3. Sunset, 4. Night.

Reference only. Do not write on this sheet. Keep the Sun fixed on the left and follow the red marker anticlockwise (1 → 2 → 3 → 4).

| Position in North-Pole view | Time of day | Space viewpoint clue (what the marker does) | Observer viewpoint clue (how the sky looks) |
|---|---|---|---|
| Position 1 · Top boundary | Sunrise | Turns anticlockwise from the unlit half into the sunlit left half | Sun appears low in the eastern sky |
| Position 2 · Left sunlit side | Middle of the day | Faces directly towards the fixed Sun on the left | Sun appears high across the sky |
| Position 3 · Bottom boundary | Sunset | Turns anticlockwise out of the sunlit half into the unlit right half | Sun appears low in the western sky |
| Position 4 · Right unlit side | Night | Faces away from the fixed Sun inside the shaded right half | Sky is dark; the Sun is below the horizon |

## Reference B · Key Stage 3: tilted axis (23.5°), day length, seasons, and time zones




Key Stage 3 model: Earth's 23.5° tilted axis causes changing day length and seasons, while 360° rotation in 24 hours (15° per hour) causes time zones.

| Key Stage 3 concept | Space mechanism (actual cause) | Observable effect on Earth |
|---|---|---|
| Apparent Sun motion | Earth rotates west to east on its axis once every 24 hours | The observer turns with Earth, so the Sun appears to move from east to west across the sky |
| Time zones (15° per hour) | Earth rotates 360° in 24 hours, turning 15° of longitude each hour (360° ÷ 24 = 15°) | Global time zones shift by 1 hour for every 15° of longitude east or west |
| Tilted axis & length of day | Earth's axis is tilted at 23.5° relative to its orbit around the Sun | Daylight is longer than 12 hours when our hemisphere tilts towards the Sun and shorter when it tilts away |
| Seasons | Earth orbits the Sun once a year while keeping its 23.5° axial tilt pointed in the same direction | Hemisphere tilted towards the Sun has summer (longer days, higher Sun); opposite hemisphere has winter (shorter days, lower Sun) |

**Made by Matt**

# Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

## Arrival · 4 questions

**1.** Use the arrival picture cue. As Earth rotates anticlockwise in this North-Pole view, what happens as a marker at the top turns into the left half?

*Help: Follow the anticlockwise arrow from the top boundary into the sunlit left half.*

**2.** Earth rotates 360° in 24 hours. Through how many degrees of longitude does Earth turn in 1 hour, and how does this cause time zones?

*Help: Divide 360° by 24 hours to link rotation speed to time zones.*

**3.** At what angle is Earth's axis tilted, and how does this tilted axis affect the length of day through the year?

*Help: Link the 23.5° axial tilt to how long a place stays in the lit half during one rotation.*

**4.** How does Earth's tilted axis explain why the Northern and Southern Hemispheres have opposite seasons?

*Help: Compare which hemisphere tilts towards the Sun and which tilts away at the same point in Earth's orbit.*

## You do tasks

**1. Explain how Earth's west-to-east rotation causes both the apparent east-to-west motion of the Sun and global time zones (using 360° ÷ 24 hours = 15° per hour).**

**Staff I ☐ S ☐ C ☐**

*Help: Separate the observer's apparent sky view from Earth's actual rotation, then calculate the longitude turned in 1 hour.*

**2. Explain how Earth's 23.5° tilted axis changes the length of day through the year between the summer solstice, winter solstice, and equinoxes.** **Staff I ☐ S ☐ C ☐**

*Help: Use Reference B to compare how much of a latitude circle lies inside the sunlit half when tilted towards vs away from the Sun.*

**3. Use the tilted axis (23.5°) and Earth's orbit to explain the seasons, and state why four equal quarter turns on a non-tilted model only represent an equinox.** **Staff I ☐ S ☐ C ☐**

*Help: Link axial tilt to both daylight length and the angle of sunlight hitting the surface.*

**Optional participation-loop prompt:** Share one explanation with an adult, hear one feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

## Exit ticket

**1.** Explain Place X's night and later daylight using both the space viewpoint (rotation) and the Earth observer's viewpoint (apparent movement).

**2.** How does Earth's 360° rotation in 24 hours (15° per hour) explain time zones?

**3.** How does Earth's tilted axis (23.5°) explain changing length of day through the year and the seasons?

**Pupil voice**

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

## Feedback and response · Aspire

### Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

| Code | Meaning |
|---|---|
| VF | Verbal feedback |
| WS | Worked with support |
| I | Independent |
| NS + … | Next step |
| E | Evidence captured |
| R | Pupil responded to feedback |
| // | Self-edit point |
| ? | Read this back to me |

### My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

**Made by Matt**